

Question Number	Answer	Additional guidance	Mark
6(a)(i)	role of an organism in its {environment / habitat / ecosystem} (1)	accept species/octopus etc for organism ignore animal ACCEPT niche includes all of the abiotic and biotic aspects of the (organisms) habitat (1)	(1)

Question Number	Answer	Additional guidance	Mark
6(a)(ii)	anatomical / physiological (1)	accept phonetic spellings accept anatomic ignore physical	(1)

Question Number	Answer	Additional guidance	Mark
6(a)(iii)	An answer that makes reference to the following points: (looking like a flat fish) therefore {more likely to catch prey / reduce risk of predation / camouflaged} (1) (looking like a venomous sea snake) reduced predation / less likely to be eaten / predators would avoid venomous prey (1)	accept {predators / prey} will not {see / recognise} them / hide from predators / blend into environment / lure prey / enter flat openings accept they can get among flat fish to prey on them ignore reduce {competition / predation} from flat fish ignore hide unqualified accept protection from {predators / sea snakes} increase survival chance accept scare predators away / deter predators / look like a predator instead of prey ignore defending themselves with venom/eq General Marking Guidance All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.	(2)

		• Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions. • Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie. • There is no ceiling on achievement. All marks on the mark scheme should be used appropriately. • All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer	
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		matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme. • Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited. • When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted. • Crossed out work should be marked	
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		UNLESS the candidate has replaced it with an alternative response.	
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Question Number	Answer	Mark
6(b)(i)	The only correct answer is A control <i>B is not correct because the current pain-relieving drug is used as a control</i> <i>C is not correct because the current pain-relieving drug is used as a control</i> <i>D is not correct because the current pain-relieving drug is used as a control</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(b)(ii)	A description that makes reference to four of the following points: • (stage I) testing on healthy {people / volunteers} (without cancer) (1) • (stage III) testing on {1000 to 3000 / large number of} cancer patients (1) • double blind trial (1) even if in stage II • analysis of results with (appropriate) statistical test / test for significant difference / review by independent {scientists / medics/FDA} (to see if work can progress to next stage) (1) • identification of appropriate {concentrations / dosage} (1)	accept computer modelling accept testing on {human cells / human tissues / animals} ignore patients unqualified ignore patients with {pain/ disease} accept description of double blind accept peer review by scientists ignore {comparing results / analysis} unqualified accept {identification of / rule out} {side effects / adverse reactions / toxicity / how drug is metabolised}	(4)

Question Number	Answer	Additional guidance	Mark
6(c)	An answer that makes reference to four of the following points: - idea that both drugs provide pain relief / current drug provides 35 hours of pain relief / tetrodoxin provides 460(+) hours of pain relief / few patients experience the longest pain relief (1) - tetrodoxin more effective pain relief drug / converse (1) - consideration of subjective nature of {pain / duration of pain relief} (1) {small sample size / no repeats / no SD / no error bars / no statistical test} linked to {validity/reliability/ significance} (1) - no information on age / sex / lifestyle / type of cancer / dosage / side effects etc (1)	Piece together accept {same trend / largest decrease} {at the beginning / suitable stated time} accept linear decrease for current pain-relieving drug whereas {non-linear / exponential} decrease for tetrodoxin accept tetrodoxin {has longer period of pain relief / lasts longer / is better pain relief drug} / converse awarding mp2 means mp1 as well e.g. different people have different pain tolerance accept carried out on (only) {100 to 500} (cancer) patients / each group contained (only) {50 to 250} (cancer) patients ignore no information about sample size accept gender	(4)